

Escalate or Answer: How Should an Agentic RAG Assistant Hand Back a Decision That Depends on the User?

CS567_Coen_Cam_Context_Not_Found

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ABSTRACT

RAG assistants over internal documentation answer fluently even when the right answer depends on who is asking. Agentic RAG architectures prescribe a remedy: when the agent cannot settle a choice from the documents alone, it hands the decision to the human. How that hand-back should look is untested where the agent’s default can be wrong. We propose a within-subjects study ($N=18$ target, 12 minimum) with Colorado State University students new to a department computing system. On each trial the assistant streams a short cited procedure over a frozen documentation snapshot and pre-fills a settings card; one setting depends on a fact about the participant that the question omitted. We vary only the form of the escalation (modal stop, inline flag, provenance-only cue), holding content, provenance and undo constant, and measure source opening, whether people keep or change the agent’s default when it is right and wrong, time, and NASA-TLX.

KEYWORDS

agentic AI; retrieval-augmented generation; interruption; appropriate reliance; human-in-the-loop; proactive assistants

1 INTRODUCTION AND MOTIVATION

Retrieval-augmented generation (RAG) is the usual way to put a large language model on top of an organization’s own documents, and internal documentation is full of answers that are conditional. Which GPU partition to use depends on how much memory the model needs; how to connect depends on whether you are on campus; where a job should write depends on whether the data matters. People ask short questions that leave the deciding fact out (“Set up a GPU job for my PyTorch model”), and a RAG assistant composes a confident procedure from whichever page ranked higher. Nothing on screen says that a second page applies to someone else, and a reader who is new to the system has no way to tell.

The reliance literature gives reason to expect that the reader will not catch it. Bansal et al. [1] found that explanations raise acceptance of AI recommendations whether or not they are right. Bućinca et al. [2] showed that cognitive forcing functions reduce overreliance but are the designs people like least. Vasconcelos et al. [14] modeled verification as a cost–benefit choice: people check the AI only when checking is cheap relative to the task. The practical question is therefore not whether to show sources but how to hand the decision back at the moment it is needed.

Agentic RAG architectures describe such a moment. Mishra et al. [9] catalogue a human-in-the-loop pattern in which the agent pauses execution to request disambiguation when it cannot resolve a choice itself. Sánchez-Adame and Mendoza [12] supply a contract, Minimal Proactivity Policies (MiPP), for how a system-initiated pause should be expressed, and tested three initiation forms in a within-subjects study of $n=48$, but on a system that was never wrong, so checking the source could not change any outcome. This proposal asks what neither line has answered:

RQ. When an agentic RAG assistant finds mid-task that the right answer depends on a fact about the user that the question did not state, and hands the choice back, how does the form of that hand-back (modal stop, inline flag, or provenance-only cue) affect verification, appropriate reliance, and cost?

We hold the agent constant and vary only the human-facing form. Each trial is one assistant session over a frozen snapshot of Colorado State University (CSU) Computer Science systems documentation; the participant reviews a short procedure and a card of settings and submits the card. The escalation points are *context-dependent forks*: two pages give different answers, both correct for someone, and which applies depends on the participant’s situation, which the assistant was never told. Such a fork is exactly the case in which escalating is good agent design rather than a defect, because the human holds the deciding information. The agent’s default is sometimes wrong for this participant, so verifying has a payoff and keeping or changing the default can be scored as appropriate or inappropriate reliance [13]. The contribution is (1) the first comparison of escalation forms at fixed decision points on an agent whose default can be wrong, with verification and appropriate reliance as outcomes; (2) a pre-specified prediction of the verification–cost trade-off across forms and its test; and (3) a reusable protocol (frozen snapshot, item records, contamination filter, matched task sets, event logging) that other groups can apply to their own documentation.

2 RELATED WORK

Interruption as a design variable. McFarlane and Latorella [8] made human interruption a first-class problem for interface design. McFarlane [7] compared four coordination methods (immediate, negotiated, mediated, scheduled) in a dual-task experiment and found that they trade off differently on the interrupted task,

the interrupting task and subjective measures, with negotiated interruption performing well overall and immediate interruption favoring prompt handling.^[verify] Horvitz [5] framed mixed initiative decision-theoretically: initiate only when the expected utility of acting exceeds the cost of interrupting, and use dialogue to resolve uncertainty. Read against this work, and as our own interpretation, a modal stop is closest to McFarlane’s immediate method, an inline flag to a negotiated one, and a provenance-only answer does not interrupt at all. In these studies the interruption carried no fallible content.

Reliance on AI advice. Work at CHI, CSCW and IUI studies what people do with an AI recommendation that may be wrong. Bansal et al. [1] found no complementary gain from explanations, which raised acceptance regardless of correctness. Bućina et al. [2] showed that forcing engagement before the AI’s answer is accepted reduces overreliance at a subjective cost. Vasconcelos et al. [14] showed across five studies that explanations help only when they lower the cost of verifying relative to the task. Schemmer et al. [13] define appropriateness of reliance in two dimensions: following the AI when it is right and overriding it when it is wrong. Möhle et al. [10] found in a judge–advisor study that a stated accuracy level shifted self-reported trust more when the reasoning was convincing, while actual accuracy was never varied; we take from this that self-report is a weak proxy for behavior, which is why our primary outcomes are logged. All of these are single-shot decisions in which the AI proposes and the human disposes; none has a system-initiated pause mid-task, and none varies its form.

MiPP: a contract for proactive initiation. MiPP [12] separates decision authority (whether the system may initiate) from interaction authority (how the initiation is expressed). Its constraints include C1, a brief cue followed by a single question or choice; C2, a one-step escape hatch; and C3, an explicit provenance cue naming the source that justified the initiation. C5 says that when inputs are missing or ambiguous the system should “defer, downgrade to an indirect channel, or remain silent.” MiPP’s $n=48$ study compared a reactive baseline with three policy families (direct alert, confirmatory prompt, indirect entry) on productivity scenarios; the quieter families reduced perceived intrusion, and removing the confirmatory cue worsened intrusion, timing, trust and willingness. Three features define the gap we address. Each condition bundled a family with its own trigger and scenario, so forms were not compared at the same decision points; every initiation was correct by construction, so verification had no consequence and the families differed on it by at most six points; and its verification measure is anchored to an initiation, so it is undefined when nothing is announced. We anchor verification to the item. MiPP calls its contract domain-agnostic and names per-scenario designs with real workloads as future work.

C5 also creates a tension that our design tests. C5 prescribes silence when the system is uncertain about the *user’s context*, which is exactly our case: the agent does not know the fact that decides the fork. But silence here means quietly applying a default that is wrong for this user. The provenance-only condition is C5 taken literally.

Agentic RAG and RAG transparency. Mishra et al. [9] systematize agentic RAG and document a human-in-the-loop pattern that “models human oversight as a callable API within the action space,” pausing execution “to request disambiguation or supervision” when uncertainty exceeds a threshold. They note that existing implementations validate at predefined bottlenecks rather than triggering on internal ambiguity, and *propose* escalation precision and recall and detection of conflicting retrieved contexts as criteria for future systems; the paper is a survey, available only as an arXiv preprint, and evaluates nothing. Ravi and Singhgatta [11] evaluated a RAG assistant for equity research with financial professionals and report that source attribution and highlighting substantially improved self-reported trust and understanding, whereas confidence scores alone did not.^[verify] Their measures are self-report, the system is not agentic, and nothing interrupts mid-task; a cited answer with highlighted sources is the current practice that our provenance-only condition reproduces. Kirchenbauer et al. [6] validated FictionalQA items by prompting the generator blind and informed and discarding items answerable blind; the check reduces, not removes, the pretraining confound, and we reuse it as a filter on real items. Chakraborti et al. [3] are precedent for asking students, given the ground truth, to judge an agent’s plans; their dataset held no incorrect plans, so we claim no calibration procedure from it.

The gap. Interruption research tells us how coordination method changes disruption when the interruption is not fallible. Reliance research tells us how forcing engagement changes overreliance in single-shot decisions. MiPP tells us how an initiation should be expressed, on a system that is never wrong. No study has fixed the decision points of a fallible agent, varied only the form in which it hands a decision back mid-task, and measured whether people verify and rely appropriately, and at what cost.

3 APPROACH

3.1 The fork: a decision that depends on the user

Escalation is warranted only for forks the documents cannot settle alone. A *context-dependent fork* is one where two pages of the corpus give different answers, each correct for someone, and which applies depends on a fact about the user that the question does not state. In the CSU CS systems documentation the GPU Jobs page lists RTX 3090 GPUs with 24 GB on the `kestrel-gpu` partition and A100 GPUs with 40 or 80 GB on the `peregrine-gpu`; which partition a PyTorch job should use depends on the model’s memory need, which “Set up a GPU job for my PyTorch model” does not say. A competent agent cannot resolve that from the documents; a human who knows their own situation can. This is Mishra et al.’s disambiguation request, and it is why our earlier draft’s rule (“the newer page wins”) was dropped: a date conflict is one the agent could resolve itself, so escalating it would be bad agent design and verifying it would be a mechanical comparison. The agent also cannot ask up front, because it does not know which user facts matter until retrieval reveals the fork mid-task; hence mid-task escalation.

The task card is therefore split. *Your situation* is one or two lines only the participant sees (“Your PyTorch model needs about 30 GB

of GPU memory”). *What you asked the assistant* is a short question, and the assistant receives only that, as people actually ask. The escalation names the options but not their conditions (“Two pages give different GPU options: kestrel-gpu (RTX 3090) or peregrine-gpu (A100). I’ll use kestrel-gpu”), so in every form deciding requires opening a page and checking it against one’s own situation. The key for an item is derived from the corpus page and the situation line alone: no dates, no authority rule, no verification against the live system. The default is the higher-ranked retrieval option. Default-correct items reuse the same pages with a different situation line (a model that fits in 16 GB), so a participant who changes the default there is switching inappropriately, which separates calibration from blanket distrust.

3.2 One trial

An item takes 60–90 s and participants never type. (1) The task card appears. (2) The assistant pane streams three or four short cited steps at about 3 s per step with status lines (“Searching the cluster documentation...”); on flagged items exactly one step is the fork. (3) A *run card* shows three or four fields, all dropdowns with three or four options drawn from the corpus, pre-selected with the assistant’s choices. The scored field’s options include both diverging options and one or two plausible alternatives; the other fields are correct, uncontested decoys, so the card does not point at the fork, which would announce it in every condition. (4) The participant submits; no feedback is given. Scoring is automatic: scored field equals key. Per trial we obtain the submitted card and the event log, from which we derive accuracy, the reliance decision (kept vs. changed the default, crossed with default correctness), *verified* (opened a diverging page before submitting) and time. We do not ask “is the assistant correct?”: that would announce the fork in every condition, and people act on output rather than grade it. Control items have no fork; some of their steps cite two agreeing pages, so two citation chips never signal a fork, and they give a baseline page-opening rate.

3.3 The three forms

All three forms share one skeleton: the same steps, the same default, the same two pages as citations, the same controls (use the other option, open a page) with one-step undo. This holds MiPP’s C1–C3 constant. The forms are defined by two properties, whether the fork is *announced* and whether the task is *blocked*: **modal stop** (announced, blocking) pauses streaming at the fork step, overlays the sentence with buttons for both options and *Open both pages*, and disables the card until a choice is made; **inline flag** (announced, not blocking) streams all steps and puts the same sentence with *Use peregrine-gpu* and *Open pages* on the fork step, card pre-filled, ignorable; **provenance-only** (not announced) streams all steps, the fork step reads “Use partition kestrel-gpu (RTX 3090)” with two citation chips and no warning, and the fork is discoverable only by opening a page. Figure 1 shows all three for one item. The forms are not MiPP’s families, which also differ in trigger and in whether the system acts before or after asking; we take MiPP’s constraints, not its taxonomy. Provenance-only is current practice [11] and the C5 silence of Section 2. Verification is defined

as opening a diverging page, which no form forces, so a difference cannot be produced by the modal form requiring a click; the modal form does force a *choice*, which is inherent to blocking and reported as such.

3.4 Why this is an agentic, human-centered study

Items are drafted by a minimal agentic RAG loop (Figure 2): given the question it plans sub-steps, retrieves the top-*k* chunks per sub-step from the snapshot, and runs a fork check that fires when the retrieval-score margin between the top candidates is small *and* an LLM judge answers yes to “do these chunks give different answers depending on a user attribute the question does not state?” Only items on which the check fired become flagged items; we report its hit rate. Each item is then frozen as a JSON record (situation, question, steps with citations, fork step index, options, default, card fields with options and pre-fill, scored field, key), and the web app reads only the record, so every participant sees identical content and the condition flag changes only the fork step’s rendering.

Pre-generated, replayed behavior. Agent behavior is generated by the pipeline before the sessions and replayed identically to every participant, following the fixed-advice paradigm common in AI-reliance studies [2, 13, 14]^[verify] and the older practice of controlling system behavior in intelligent-interface studies [4]^[verify], so that escalation points, defaults and default correctness are controlled. The pipeline runs locally on an open-weight model at temperature 0 with a fixed seed; snapshot, records and model digests are released. Authors may edit wording but never an escalation decision or a default; default correctness is set by the situation line. Participants are told responses were prepared in advance. The study thus isolates the human-facing form of an agent-initiated escalation, not the agent’s behavior or multi-turn interaction. This passes the “if a chatbot can do it” test: a chatbot answers the question as asked, and only a loop that retrieves per step and inspects its own retrieval can discover mid-task that the answer depends on an unstated fact and choose to hand the decision back. The human-centered question is what that hand-back should look like, and every outcome is a human behavior. “Agentic” rests on four properties, each verifiable in the released records: the task is multi-step, the escalation arrives at the affected step, the trigger is the loop’s own decision, and the loop is fully described here.

3.5 Corpus and items

The corpus is drawn from the CSU CS department systems documentation (sna.cs.colostate.edu); which sections are used is not yet decided. Requirements are fixed: a frozen snapshot; every key derivable from the snapshot plus the situation line, with no administrator verification; snapshot and item records released together; 18 measured items (12 flagged, 6 control) plus 2 practice items from three disjoint sections (for example GPU and jobs, access and remote connection, storage and software); and a system many CSU students have not used. Table 1 gives illustrative items written against the current pages. We keep the FictionalQA blind-versus-informed check [6] as a filter: the drafting model is queried blind and with the two pages, and facts it answers correctly blind are discarded (department-local hostnames, partition names and paths

(a) Modal stop: announced, blocking

YOUR SITUATION
CS graduate student. Your PyTorch model needs about 30 GB of GPU memory.

WHAT YOU ASKED
"Set up a GPU job for my PyTorch model."

RUN CARD

Partition
kestrrel-gpu ▾

QoS
gpu_short ▾

GPUs
1 ▾

Module
python/anaconda ▾

Submit (disabled)
waiting for your choice

ASSISTANT
Searching the cluster documentation...

1. Connect: ssh falcon.cs.colostate.edu
[Connecting to Falcon](#)

2. Choose a GPU partition...
Paused at step 2.

Two pages give different GPU options: kestrrel-gpu (RTX 3090) or peregrine-gpu (A100). I'll use kestrrel-gpu.

[Use kestrrel-gpu](#)
[Use peregrine-gpu](#)
[Open both pages](#)

One-step undo after choosing.

(b) Inline flag: announced, not blocking

YOUR SITUATION
CS graduate student. Your PyTorch model needs about 30 GB of GPU memory.

WHAT YOU ASKED
"Set up a GPU job for my PyTorch model."

RUN CARD

Partition
kestrrel-gpu ▾

QoS
gpu_short ▾

GPUs
1 ▾

Module
python/anaconda ▾

Submit

ASSISTANT
Searching the cluster documentation...

1. Connect: ssh falcon.cs.colostate.edu
[Connecting to Falcon](#)

2. Use partition kestrrel-gpu (RTX 3090) [GPU Jobs](#) [Partitions](#)

Two pages give different GPU options: kestrrel-gpu (RTX 3090) or peregrine-gpu (A100). I'll use kestrrel-gpu.

[Use peregrine-gpu](#)
[Open pages](#)

3. Job script: --qos=gpu_short, module load python/anaconda
[PyTorch](#) [Partitions](#)

4. Submit: sbatch job.sh
[Submitting Jobs](#)

Done. Review the run card and submit.

(c) Provenance-only: not announced

YOUR SITUATION
CS graduate student. Your PyTorch model needs about 30 GB of GPU memory.

WHAT YOU ASKED
"Set up a GPU job for my PyTorch model."

RUN CARD

Partition
kestrrel-gpu ▾

QoS
gpu_short ▾

GPUs
1 ▾

Module
python/anaconda ▾

Submit

ASSISTANT
Searching the cluster documentation...

1. Connect: ssh falcon.cs.colostate.edu
[Connecting to Falcon](#)

2. Use partition kestrrel-gpu (RTX 3090) [GPU Jobs](#) [Partitions](#)

3. Job script: --qos=gpu_short, module load python/anaconda
[PyTorch](#) [Partitions](#)

4. Submit: sbatch job.sh
[Submitting Jobs](#)

Done. Review the run card and submit.

Figure 1: Item G1 at the fork under the three forms (wireframes). In each panel the left column holds the task card (*Your situation*, seen only by the participant; *What you asked*, the only input the assistant received) and the run card (dropdowns pre-selected with the assistant’s choices; only Partition is scored). The center is the assistant pane; a source panel that opens a cited page with the passage highlighted sits to its right and is omitted here. Steps, citations, default and undo are identical; only the rendering of step 2 differs.

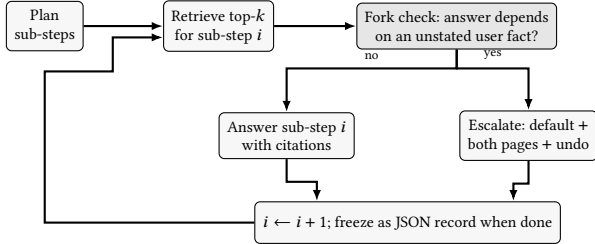


Figure 2: The loop that drafts each item. The fork check is the retrieval-score margin plus an LLM judge. Its output is frozen and replayed identically to every participant; only the rendering of the “Escalate” step differs between conditions.

are the kind that fail blind). Task sets are matched on number of steps, number of card fields, question–page lexical overlap band, fork type and fork position, checked in the pilot, and item is a random effect in every model. Because someone who knows that a 3090 has 24 GB could resolve G1 without opening a page, we screen for prior use, prefer items whose condition is stated only on the page, and log what was opened.

4 SYSTEM VISION AND TECHNOLOGY

The apparatus is one web application (React or plain JavaScript) with a condition flag, run in a desktop browser with no headset or spatial component. The screen has three panes: task card and run card on the left, the assistant pane in the center, and a source panel

Table 1: Illustrative items from the CSU CS systems documentation. Only the scored field is compared with the key. G2 is default-correct; C1 is a control item. Page facts are from the current pages; the exact pairing is illustrative until the snapshot is frozen.

ID	Your situation / what you asked	Pages and options (scored field)	Default → key
G1	Model needs about 30 GB of GPU memory. "Set up a GPU job for my PyTorch model."	GPU Jobs, Partitions: kestrrel-gpu (RTX 3090, 24 GB each) vs. peregrine-gpu (A100, 40/80 GB). Field: partition.	kestrrel-gpu (wrong) → peregrine-gpu
G2	Model fits in 16 GB. Same question.	Same pages and field.	kestrrel-gpu (correct)
R1	Working from home. "How do I connect to Falcon?"	Connecting to Falcon: ssh falcon.cs.colostate.edu; SSH guide: off campus, connect to the CSU VPN first. Field: first step.	ssh directly (wrong) → VPN, then ssh
S1	Output is scratch data you can regenerate. "Where should my job write its output?"	Home Directories: home (backed up, quota); Temporary Disk Space: /s/<host>/a/tmp (not backed up, scrubbed). Field: output path.	home (wrong) → /s/<host>/a/tmp
C1	(none) "Why is my job still pending?"	FAQ and Monitoring Jobs agree: squeue, REASON column. No fork. Field: command.	squeue (correct)

on the right that opens a cited page with the relevant passage highlighted (Figure 1). A replay engine reads the item’s JSON record and streams it at a fixed pace; the flag selects one of three renderings of the fork step and nothing else. Every interaction is logged as a timestamped event (item, condition, fork onset, each control used, each page opened and for how long, each dropdown change, the submitted card, block boundaries) to a local SQLite database under a participant code. Instruments appear as full-screen steps

between blocks. The drafting pipeline is Python: the snapshot is chunked by heading, embedded into a vector index, and queried per sub-step; one LLM plans sub-steps and acts as the fork judge. The model will be named in the paper; because records are frozen, the human-facing study does not depend on it. We estimate one week for the pipeline and two for the web application.

5 EVALUATION PLAN

Design and participants. Independent variables: escalation form (modal, inline, provenance-only), within subjects; and default correctness at flagged items (correct, wrong), within block. A 3×3 Latin square of task set and form with all six orders; $N=18$ target (three per order), 12 minimum; pilot of 3–4. A between-subjects design would need roughly three times as many people for the same power, which at class scale leaves four to six per condition, so we keep within subjects. Participants are CSU students who have not used the documented system (screening item), recruited through the class data-collection form; the realistic pool is upper-division undergraduates and new graduate students. The class form suffices for the course; if we intend to publish, an IRB protocol with a faculty principal investigator is needed before recruitment, and we will raise it with the instructor in the first two weeks of October.

Procedure. A session takes about 45 minutes: consent and instructions, which tell everyone that the assistant is usually right but can be wrong (5 min); a two-item practice block without forks (3 min); three measured blocks of six items (4 flagged, 2 default-correct and 2 default-wrong, plus 2 control; order randomized within block), each followed by raw NASA-TLX and one perceived-intrusion item (about 9 min per block); and a closing questionnaire with a manipulation check (identify the form just used), a ranking of the three forms and a free-text reason (5 min). Every critical fact appears in exactly one item and task sets come from disjoint sections; generic steps such as connecting recur but are never scored. Block position enters the models as a covariate and the first block is reported alone as a descriptive between-subjects check.

Hypotheses and measures. **H1 (verification):** on flagged items the proportion of items on which a diverging page is opened before submission is ordered modal > inline > provenance-only. **H2 (appropriate reliance):** correcting a wrong default is higher under modal and inline than under provenance-only, while keeping a correct default does not differ, i.e., a form $\times$ default-correctness interaction on the scored decision. **H3 (cost):** time per flagged item and NASA-TLX per block are ordered modal > inline > provenance-only. H1 and H3 together are the trade-off reported for forcing functions [2]; H2 is what matters to a designer. Measures are: verified; the scored-field decision crossed with default correctness, from which relative AI reliance and relative self-reliance follow [13]; time from item onset to submission; raw NASA-TLX per block; one perceived-intrusion item per block (1–7, as in MiPP-Eval, for comparability); the manipulation check; and the ranking with its reason. Nothing else is collected. The ranking is reported descriptively.

Analysis. Primary analyses are at the trial level (four flagged items per condition per participant): logistic mixed models for verified and for the scored decision, with fixed effects of form,

default correctness, their interaction and block position, and random intercepts for participant and item; a linear mixed model on log time; repeated-measures ANOVA with Greenhouse–Geisser correction, or Friedman, for TLX and intrusion. Planned contrasts (modal vs. inline, inline vs. provenance-only) are Holm-corrected and reported as odds ratios or d_z with 95% CIs. Free-text reasons are coded thematically by both authors. A paired t -test at $N=12$ detects only large participant-level effects ($d_z \approx 0.9$ at 80% power), which is why trial-level models are primary; we will simulate power for the H2 interaction from pilot estimates.

Threats. Strategy carry-over (learning that pages can disagree) is addressed by the up-front instruction, all six orders, the position covariate and the first-block check. Domain knowledge that resolves a fork without opening a page is addressed by screening, item selection and logs. Forced choice under modal is inherent to blocking; verification is a never-forced act. Generality is limited to one corpus, novices, a short lab session and replayed single-turn sessions without follow-up questions, and is stated as a scope condition.

6 DELIVERABLES

By mid-November: the frozen snapshot of the chosen sections; the drafting pipeline with its fork check and a log of where it fired over about 30 candidate questions; the contamination-filter survivor count; 18 measured and 2 practice item records assembled into three matched task sets; the web application with the three renderings, replay engine, instruments and event logging; a completed pilot of 3–4 people with items and timing revised; and the IRB decision. **By the early-December presentations:** 12–18 completed sessions; the analysis scripts and results for H1–H3; a paper draft with Method and Results; a demo video of one item in all three forms; and the released snapshot plus item records. The critical path is item authoring, not code.

Team. Group CS567_Coen_Cam_Context_Not_Found. Coen Petto: study design, web application and participant sessions. Cameron Suess: drafting pipeline, item authoring and corpus snapshot. Both authors write and analyze.

7 CONCLUSION

We propose a within-subjects study of how an agentic RAG assistant should hand back a decision that depends on a fact about the user it was never told. Only the form varies. The prediction is that a modal stop buys the most verification at the highest cost, that an inline flag captures most of the benefit for less, and that a provenance-only cue, which is what current RAG interfaces provide and what MiPP’s C5 counsels, leaves the agent’s errors largely uncaught.

USE OF AI

Claude Code (Anthropic), running Claude Opus 5.5 with Claude Fable 5.1 subagents, was used to critique the proposal, check claims against MiPP, check references against publisher records, draft this text and draw the mock-ups. The authors made every design decision, revise the text and verify each reference; claims still to be checked are marked [\[verify\]](#).

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